

Homework 5

Tiffany Tu

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Setup

This Quarto document contains only runnable R code and display-math (no inline math) and assumes the following CSV files are present in the same directory as this .qmd:

- trading_behavior.csv
- yogurt_data.csv

All math expressions are placed in display mode to ensure LaTeX safety.

```
# Packages
if(!require(mlbench)) install.packages("mlbench", repos='https://cloud.r-project.org')
if(!require(MASS)) install.packages("MASS", repos='https://cloud.r-project.org')
library(mlbench)
library(MASS)
```

Question 1: Linear Regression with Normal Errors

Model and log-likelihood (display math)

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + e_i$$

$$e_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - X_i' \beta)^2$$

The canonical link is the identity link. The inverse link is the identity function.

```
# Load BostonHousing2 and construct matrices
data("BostonHousing2", package = "mlbench")
bh <- BostonHousing2
y <- bh$medv
X <- as.matrix(cbind(1, bh$rm, bh$age, bh$crim))
colnames(X) <- c("Intercept", "rm", "age", "crim")
n <- nrow(X); k <- ncol(X)
```

R function: Normal log-likelihood

```
loglik_normal <- function(theta, X, y) {
  k <- ncol(X)
  beta <- theta[1:k]
  sigma2 <- theta[k+1]
  if(!is.finite(sigma2) || sigma2 <= 0) return(-1e12)
  n <- length(y)
  res <- as.numeric(y - X %*% beta)
  ll <- - (n/2) * log(2*pi*sigma2) - (1/(2*sigma2)) * sum(res^2)
  return(ll)
}
```

MLE via optim() and SEs from Hessian

```

init_theta <- c(rep(0, k), var(y))
nll <- function(theta, X, y) -loglik_normal(theta, X, y)
opt_norm <- optim(par = init_theta, fn = nll, X = X, y = y, hessian = TRUE, method = "BFGS",
theta_hat <- opt_norm$par
beta_hat <- theta_hat[1:k]
sigma2_hat <- theta_hat[k+1]
H_norm <- opt_norm$hessian
vcov_norm <- try(solve(H_norm), silent = TRUE)
se_theta <- if(inherits(vcov_norm, "try-error")) rep(NA, length(theta_hat)) else sqrt(diag(v
# Print estimates
beta_hat; sigma2_hat; se_theta

```

```
[1] -23.60962629  8.03342603 -0.05223763 -0.21102448
```

```
[1] 36.8445
```

```
[1] 2.75865452 0.40044939 0.01042223 0.03393366 2.31680148
```

Compare to lm()

```

lm_mod <- lm(medv ~ rm + age + crim, data = bh)
summary(lm_mod)

```

Call:

```
lm(formula = medv ~ rm + age + crim, data = bh)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-19.959	-3.143	-0.633	2.150	39.940

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-23.60556	2.76938	-8.524	< 2e-16 ***
rm	8.03284	0.40201	19.982	< 2e-16 ***
age	-0.05224	0.01046	-4.993	8.21e-07 ***
crim	-0.21102	0.03407	-6.195	1.22e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.094 on 502 degrees of freedom

Multiple R-squared: 0.5636, Adjusted R-squared: 0.561

F-statistic: 216.1 on 3 and 502 DF, p-value: < 2.2e-16

Bootstrap SEs for beta (nonparametric)

```
set.seed(123)
B <- 2000
coef_mat <- matrix(NA, nrow = B, ncol = k)
colnames(coef_mat) <- c("Intercept", "rm", "age", "crim")
for(b in 1:B) {
  idx <- sample(1:n, n, replace = TRUE)
  bh_b <- bh[idx, ]
  fit_b <- lm(medv ~ rm + age + crim, data = bh_b)
  coef_mat[b, ] <- coef(fit_b)
}
boot_se <- apply(coef_mat, 2, sd, na.rm = TRUE)
boot_se
```

Intercept	rm	age	crim
4.46004359	0.68214977	0.00936242	0.03620821

Question 2: Poisson Model for Count Data (trading_behavior)

Poisson model and log-likelihood (display math)

$$Y_i \sim \text{Pois}(\mu_i)$$

$$\log(\mu_i) = X_i' \beta$$

$$\ell(\beta) = \sum_{i=1}^n [-\exp(X_i' \beta) + y_i X_i' \beta - \log(y_i!)]$$

```
# Read trading data
trade <- read.csv("trading_behavior.csv", stringsAsFactors = FALSE)
# Attempt to standardize common column names
if(!"numtrades" %in% names(trade)) {
  possible <- grep("numtr", names(trade), value = TRUE, ignore.case = TRUE)
  if(length(possible)==1) names(trade)[names(trade)==possible] <- "numtrades"
}
if(!"finlittest" %in% names(trade)) {
  possible2 <- grep("finlit|finl|score", names(trade), value = TRUE, ignore.case = TRUE)
  if(length(possible2)>=1) names(trade)[names(trade)==possible2[1]] <- "finlittest"
}
trade$program <- as.character(trade$program)
trade$numtrades <- as.numeric(trade$numtrades)
trade$finlittest <- as.numeric(trade$finlittest)
trade <- trade[!is.na(trade$numtrades) & !is.na(trade$finlittest) & !is.na(trade$program), ]
str(trade); head(trade)
```

```
'data.frame': 200 obs. of 4 variables:
 $ id      : int  45 108 15 67 153 51 164 133 2 53 ...
 $ numtrades : num  0 0 0 0 0 0 0 0 0 0 ...
 $ program  : chr  "MFE" "MSBA" "MFE" "MFE" ...
 $ finlittest: num  66 66 69 67 65 67 71 65 58 71 ...
```

	id	numtrades	program	finlittest
1	45	0	MFE	66
2	108	0	MSBA	66
3	15	0	MFE	69
4	67	0	MFE	67
5	153	0	MFE	65
6	51	0	MSBA	67


```
table(trade$program)
```

```
MBA  MFE  MSBA
105   50   45
```

R function: Poisson log-likelihood

```
loglik_poisson <- function(beta, X, y, return_vector = FALSE) {
  eta <- as.numeric(X %*% beta)
  mu <- exp(eta)
  ll_i <- dpois(y, lambda = mu, log = TRUE)
  if(return_vector) return(ll_i)
  sum(ll_i)
}
```

MLE via optim()

```
trade$MBA <- ifelse(trade$program == "MBA", 1, 0)
trade$MFE <- ifelse(trade$program == "MFE", 1, 0)
X_pois <- as.matrix(cbind(1, trade$MBA, trade$MFE, trade$finlittest))
y_pois <- trade$numtrades
init_beta <- rep(0, ncol(X_pois))
negll_pois <- function(beta, X, y) -loglik_poisson(beta, X, y)
opt_pois <- optim(par = init_beta, fn = negll_pois, X = X_pois, y = y_pois, hessian = TRUE,
beta_hat_pois <- opt_pois$par
vcov_pois <- try(solve(opt_pois$hessian), silent = TRUE)
se_pois <- if(inherits(vcov_pois, "try-error")) rep(NA, length(beta_hat_pois)) else sqrt(diag(vcov_pois))
data.frame(estimate = beta_hat_pois, se = se_pois, z = beta_hat_pois/se_pois, p = 2*(1-pnorm(abs(z))))
```

	estimate	se	z	p
1	-6.8385058	0.8917301	-7.6688071	1.731948e-14
2	1.0859384	0.3560795	3.0497080	2.290639e-03
3	0.3604895	0.4389272	0.8212968	4.114772e-01
4	0.0682507	0.0105377	6.4768103	9.368195e-11

Compare with glm()

```
glm_mod <- glm(numtrades ~ MBA + MFE + finlitest, family = poisson(link = "log"), data = tr  
summary(glm_mod)
```

Call:

```
glm(formula = numtrades ~ MBA + MFE + finlitest, family = poisson(link = "log"),  
    data = trade)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.00093	0.89706	-7.804	5.98e-15 ***
MBA	1.08386	0.35825	3.025	0.00248 **
MFE	0.36981	0.44107	0.838	0.40179
finlitest	0.07015	0.01060	6.619	3.63e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 287.67 on 199 degrees of freedom
Residual deviance: 189.45 on 196 degrees of freedom
AIC: 373.5

Number of Fisher Scoring iterations: 6

```
cbind(optim = beta_hat_pois, glm = coef(glm_mod))
```

	optim	glm
(Intercept)	-6.8385058	-7.0009343
MBA	1.0859384	1.0838591
MFE	0.3604895	0.3698092
finlitest	0.0682507	0.0701524

Likelihood Ratio Test (joint test that MFE and finlitest coefficients equal zero)

```

ll_full <- loglik_poisson(beta_hat_pois, X_pois, y_pois)
beta_constr <- beta_hat_pois
# ordering: Intercept, MBA, MFE, finlittest
beta_constr[3] <- 0 # MFE
beta_constr[4] <- 0 # finlittest
ll_constr <- loglik_poisson(beta_constr, X_pois, y_pois)
LRn <- 2*(ll_full - ll_constr)
pval_lr <- 1 - pchisq(LRn, df = 2)
data.frame(LR_stat = LRn, p_value = pval_lr)

```

```

      LR_stat p_value
1 1225.882      0

```


Question 3: Multinomial Logit (MNL) on Yogurt Data (Optional)

MNL model (display math)

$$P_i(j) = \frac{\exp(x'_{ij}\beta)}{\sum_{k=1}^J \exp(x'_{ik}\beta)}$$

```
# Read yogurt data
yog <- read.csv("yogurt_data.csv", stringsAsFactors = FALSE)
# Check columns
req_cols <- c(paste0("y",1:4), paste0("f",1:4), paste0("p",1:4))
missing <- setdiff(req_cols, names(yog))
if(length(missing)>0) stop("Missing columns in yogurt data: ", paste(missing, collapse=", "))
n <- nrow(yog); J <- 4
# Build long format
long_list <- lapply(1:J, function(j) {
  data.frame(id = yog$id,
             choice = as.numeric(yog[[paste0("y", j)]]),
             featured = as.numeric(yog[[paste0("f", j)]]),
             price = as.numeric(yog[[paste0("p", j)]]),
             prod = j)
})
long <- do.call(rbind, long_list)
long$D1 <- ifelse(long$prod==1,1,0)
long$D2 <- ifelse(long$prod==2,1,0)
long$D3 <- ifelse(long$prod==3,1,0)
long <- long[order(long$id, long$prod), ]
head(long)
```

	id	choice	featured	price	prod	D1	D2	D3
1	1	0	0	0.108	1	1	0	0
2431	1	0	0	0.081	2	0	1	0
4861	1	0	0	0.061	3	0	0	1
7291	1	1	0	0.079	4	0	0	0
2	2	0	0	0.108	1	1	0	0
2432	2	1	0	0.098	2	0	1	0

R function: MNL log-likelihood and estimation via `optim()`

```

mnl_loglik <- function(beta, long_df) {
  beta <- as.numeric(beta)
  b1 <- beta[1]; b2 <- beta[2]; b3 <- beta[3]; bf <- beta[4]; bp <- beta[5]
  util <- b1*long_df$D1 + b2*long_df$D2 + b3*long_df$D3 + bf*long_df$featured + bp*long_df$price
  expU <- exp(util)
  denom <- ave(expU, long_df$id, FUN = sum)
  prob <- expU / denom
  ll_i <- long_df$choice * log(prob + 1e-12)
  sum(ll_i, na.rm = TRUE)
}

init_beta <- c(0.2,0.2,-0.2,0.1,-5)
neg_mnl <- function(beta, long_df) -mnl_loglik(beta, long_df)
opt_mnl <- optim(par = init_beta, fn = neg_mnl, long_df = long, method = "BFGS", hessian = TRUE)
beta_hat_mnl <- opt_mnl$par
vcov_mnl <- try(solve(opt_mnl$hessian), silent = TRUE)
se_mnl <- if(inherits(vcov_mnl,"try-error")) rep(NA, length(beta_hat_mnl)) else sqrt(diag(vcov_mnl))
data.frame(estimate = beta_hat_mnl, se = se_mnl, z = beta_hat_mnl/se_mnl, p = 2*(1-pnorm(abs(z))))

```

	estimate	se	z	p
1	1.3903950	0.08813217	15.776247	0.000000e+00
2	0.6437239	0.05448292	11.815150	0.000000e+00
3	-3.0890894	0.14496052	-21.309867	0.000000e+00
4	0.4864864	0.11996084	4.055376	5.005368e-05
5	-37.1622160	2.40257128	-15.467685	0.000000e+00

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